

APAFNet: Single-Frame Infrared Small Target Detection by Asymmetric Patch Attention Fusion

Ziming Wang¹, Jianwei Yang¹, Zongxu Pan¹, Yuhan Liu¹, Bin Lei, and Yuxin Hu

Abstract—Single-frame infrared small target (SIRST) detection has been applied in many civilian and military applications and has greatly developed since deep-learning methods grow in recent years. However, the contradiction between semantic information and spatial details limits detection performance, especially for dim and small targets with complex interference. To overcome the restrictions, we propose an asymmetric patch attention fusion network (APAFNet) in this letter to merge high-level semantics and low-level spatial details, which consists of an APAF module based on a patch channel attention branch and a dilation context block, guiding the network to collect local semantics and spatial details. The experimental results on the NUAA-SIRST dataset andIRSTD-1k dataset show that the proposed APAFNet can achieve excellent performance under complex backgrounds.

Index Terms—Asymmetric patch attention fusion (APAF), cross-layer features, local semantics, single-frame infrared small target (SIRST) detection.

I. INTRODUCTION

WITH the strong ability to penetrate obstructions, anti-interference, and all-time working [1], infrared small target detection has been applied in various fields, such as fire prevention, maritime rescue, and missile guidance. However, with the development of infrared sensor platforms, the quickly changing backgrounds with moving targets will constrain the performance of most infrared small target detection methods for sequences [2]. Thus, single-frame infrared small target (SIRST) detection has attracted more attention. Nevertheless, due to the long imaging distance of the infrared detector, the target usually occupies a small region, providing insufficient texture and shape information. Meanwhile, the targets usually suffer from the low signal-to-noise ratio (SNR) and

low contrast and thus can be easily submerged into clutter backgrounds, making SIRST detection remain challenging.

Traditional methods for SIRST detection can be mainly divided into three categories: filter-based methods [3], [4], human visual system (HVS)-based approaches [5], [6], and low-rank sparse decomposition (LRSD) [7], [8], [9], [10]. Nevertheless, depending on the handcrafted features, these methods have inferior performance under various backgrounds due to the deficiency of high-level feature extraction and confusion of salient interference. Deep-learning technology, especially the convolution neural network (CNN) which can understand high-level information, has flourished in recent years and has also attracted much attention in the infrared small target detection field. To overcome the problem of small target size, Ryu and Kim designed a data-driven proposal (DDP) module to initially select possible locations of the targets [11]. Zhao et al. [12] considered the detection as an image-to-image problem and proposed a generative adversarial network (GAN) to convert the input image into a mask straightly. Moreover, Zhao et al. [13] reduced the imbalance between the backgrounds and targets by a target extraction module and a semantic constrain module. In addition to the single utilization of CNNs, the handcrafted features are also employed in [14] and [15] to construct a better representation of the infrared targets. Different from other CNN-based methods, Wang et al. [16] considered the infrared small target detection as a game between miss detection and false alarm, which is realized by two generators and one discriminator. Furthermore, Dai et al. [17] designed an asymmetric contextual modulation (ACM) module to fuse the low- and high-level features dynamically. To maintain the small targets in deep layers, [18] adopts a densely nested attention mechanism. Moreover, ISNet [19] uses a Taylor finite-difference-inspired edge block and a two-orientation attention aggregation module to precisely attain the edge of the targets. Although these existing deep-learning-based methods have achieved good performance and brought considerable developments in infrared small target detection, there are still some drawbacks.

To detect small targets under cluttered backgrounds, both spatial details and semantic information are required. Nevertheless, it is challenging for deep-learning methods to cover both, since CNNs acquire semantics by sacrificing the resolution.

To address this problem, some cross-layer feature fusion methods, like ACM [17], have been proposed to balance the requirements. In ACM, the top-down attentional modulation

Manuscript received 13 October 2022; revised 8 December 2022; accepted 12 December 2022. Date of publication 19 December 2022; date of current version 6 February 2023. This work was supported by the Aerospace Information Research Institute, Chinese Academy of Sciences. (Corresponding author: Yuhua Liu.)

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Digital Object Identifier 10.1109/LGRS.2022.3230415

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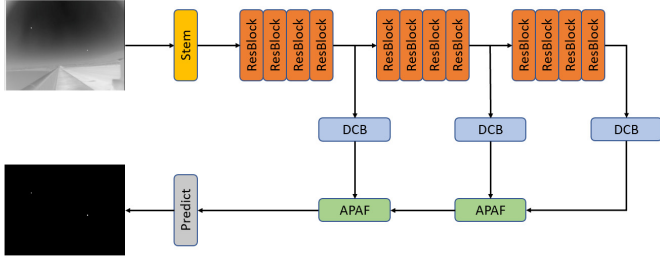


Fig. 1. Overall architecture of the network, in which APAF denotes the asymmetric patch attention fusion module and DCB means the dilation context block.

focuses on the semantic understanding of the whole image, while the bottom-up pixelwise attentional modulation pursues the pixel-level expression of the image. Nevertheless, the lack of local semantic understanding may cause the missing detection of targets that can only affect their neighboring regions. Moreover, the lack of local spatial information may blur the edge of the targets. Therefore, an asymmetric patch attention fusion network (APAFNet) is proposed in this letter. In the network, a novel cross-layer feature fusion module called APAF module is designed, which offers a special branch to collect the local semantics. Meanwhile, dilation context blocks are designed to enrich the local spatial information in the network.

The main contributions of this letter consist of the following.

- 1) The APAF module is designed for cross-layer feature fusion to emphasize the local semantics.
- 2) APAFNet built on the APAF module and dilation context block is proposed for SIRST detection.
- 3) The experiments on the NUAA-SIRST dataset andIRSTD-1k dataset verify the superior performance and robustness of our proposed network compared to other existing methods.

II. METHODOLOGY

A. Network Architecture

As depicted in Fig. 1, the proposed APAFNet, which takes an infrared image as input and outputs the segmentation result, consists of three main components: FPN as a host network, a dilation context block, and the APAF module for feature fusion operation. As the host network, FPN produces multi-level feature maps, among which high-level feature maps offer richer semantic information, while finer spatial information can be gathered from the low-level feature maps. Thus, the network exploits the spatial and semantic information of the input images more thoroughly.

A modified ResNet-20 network with one stem and three stages is used as the backbone of the host FPN to extract the feature maps from the input infrared images as shown in Table I. With only two down-sampling layers at the beginning of Stage-2 and Stage-3, the backbone can obtain higher-level semantic information without submerging fine details. In addition, the skip connections can not only pass the residuals, but also deliver high-resolution information to deep layers to some extent. Moreover, the lightweight structure with a total of

TABLE I
BACKBONE ARCHITECTURE

Stage	Output Size	Backbone
Stem	480×480	$\text{Conv3} \times 3, 16$
Stage-1	480×480	$\begin{bmatrix} \text{Conv3} \times 3, 16 \\ \text{Conv3} \times 3, 16 \end{bmatrix} \times 4$
Stage-2	240×240	$\begin{bmatrix} \text{Conv3} \times 3, 32 \\ \text{Conv3} \times 3, 32 \end{bmatrix} \times 4$
Stage-3	120×120	$\begin{bmatrix} \text{Conv3} \times 3, 64 \\ \text{Conv3} \times 3, 64 \end{bmatrix} \times 4$

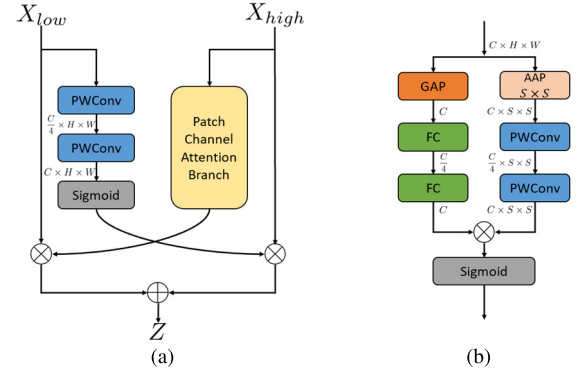


Fig. 2. Illustrations of the proposed APAF module. (a) Structure of the module, in which we use PWConv for the pixelwise convolution layer. $\otimes$ and $\oplus$ indicate multiplication and addition in pixel level, respectively. (b) Details of patch channel attention branch. GAP and FC refer to the global average pooling layer and fully connected layer, correspondingly. AAP with $S \times S$ denotes the adaptive average pooling layer with the output feature map of $S \times S$.

25 convolution layers guarantees the accuracy and efficiency for infrared small target detection.

B. APAF Module

Generally, the lower-level features contain finer spatial details, while the higher-level features provide richer semantic knowledge and these multiscale features are able to provide comprehensive information about the image. However, simply adding or concatenating the features will fuse the information in a messy way and cause a potential negative combination between the spatial details and semantic information.

To maximize the advantages of multilevel features in a clever way, an APAF module is proposed, as illustrated in Fig. 2. Two branches are designed to refine the multi-level features before fusing them. One is to highlight the high-resolution information from low-level features, while the other one aims to explore better expression of the high-level features in a semantic way.

Given a low-level feature $\mathbf{X}_{low}$ and a high-level feature $\mathbf{X}_{high}$ both with C channels and size of $H \times W$, the output of APAF module can be described as

$$\mathbf{Z} = L(\mathbf{X}_{low}) \otimes \mathbf{X}_{high} + \mathbf{X}_{low} \otimes P(\mathbf{X}_{high}) \quad (1)$$

where $\otimes$ represents elementwise multiplication. L and P represent the branches that process the low-level features and high-level features, respectively.

The subtle spatial details in large-scale low-level features are expected to be discovered because they are of concern

for the precise detection of infrared small targets. Therefore, we design two consecutive pixelwise convolution layers as a pixelwise channel attention branch, which can be accomplished via

$$L(\mathbf{X}_{\text{low}}) = \sigma(PW_2(\text{ReLU}(PW_1(\mathbf{X}_{\text{low}})))) \quad (2)$$

where σ represents the Sigmoid function. PW_1 and PW_2 are pixelwise convolution layers with batch normalization, whose kernel sizes are $(C/4) \times C \times 1 \times 1$ and $C \times (C/4) \times 1 \times 1$, respectively. We use the successive pixelwise convolution layers to construct a simple encoder-decoder for each pixel in the low-level feature. Consequently, the feature of each pixel will be aggregated on the channel dimension individually, and therefore providing highlighted details of the low-level features.

On the other hand, adequate inherent information is needed from higher-level features to detect infrared small targets. Channel attention with global average pooling is usually adopted to interchange and aggregate the information between different channels. However, global average pooling generalizes the overall feature of a channel and thus will weaken the intrinsic information of the infrared small targets. Meanwhile, the global pooling layer integrates the whole channel into one pixel, concentrating on global information while neglecting the local one, causing more attention on the most salient target naturally and potential ignorance of the weak targets when there are several infrared small targets in the images.

To address this problem, a patch channel attention branch is proposed in this work by adding another parallel patch channel attention branch with an adaptive average pooling layer to collect the local semantic information, which can be formulated as

$$P(\mathbf{X}_{\text{high}}) = \sigma(BN(W_2 \text{ReLU}(BN(W_1 \text{GAP}(\mathbf{X}_{\text{high}})))) + PW_2(\text{ReLU}(PW_1(\text{AAP}(\mathbf{X}_{\text{high}}))))) \quad (3)$$

where BN represents batch normalization and the kernel sizes of PW_1 and PW_2 are $(C/4) \times C \times 1 \times 1$ and $C \times (C/4) \times 1 \times 1$, respectively. GAP and AAP denote global average pooling and adaptive average pooling, corresponding to the different pooling strategies of the two branches. The size of the output of the adaptive average pooling is $B \times C \times S \times S$, where S is adjustable. The patch channel attention branch divides the feature into $S \times S$ patches and operates channel attention for each patch. Therefore, by changing S , the patch size, which equals the range of local regions the branch focuses, is shifted as well. Therefore, the network can not only collect global information, but also learn the knowledge of local patches, subsequently refusing interference far away from the targets and enhancing the weak targets.

C. Dilation Context Block

Although the proposed APAF module can improve the fusion results between the spatial details and semantic information of multilevel features and assist the network outline of the targets accurately, the module processes the low-level features by focusing only on single pixels. Thus, the network

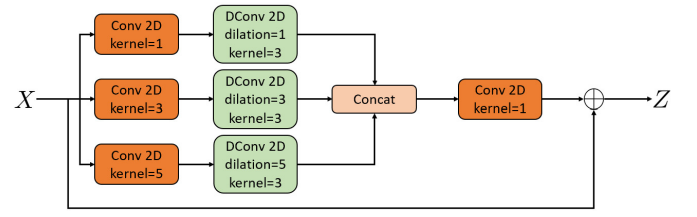


Fig. 3. Illustrations of the proposed dilation context block, in which kernel = x means the kernel size is $x \times x$. DConv 2D means dilated convolution layer.

determines the pixel to be a target only depending on the pixel itself without consideration of surrounding pixels. Under this circumstance, the edges of the targets may not be accurately decided, since the information from one pixel seems insufficient and erratic.

To better utilize the spatial context information of the pixels, we design several dilation context blocks between the backbone and the APAF modules to process the multilevel features from the backbone network. In order to thoroughly exploit the surrounding information, we adopt dilated convolution layers with different dilation rates to investigate the relations between the center pixel and its neighborhood directly, as illustrated in Fig. 3. The dilation context blocks can provide more accurate semantic information, offering precisely decisions of the contours of the targets, which can be mathematically formulated as

$$\mathbf{Z} = \mathbf{X} + \text{Conv}_{1 \times 1}(\text{Cat}[\text{Conv}_{3 \times 3, d_1}(\text{Conv}_{1 \times 1}(\mathbf{X})), \dots, \text{Conv}_{3 \times 3, d_n}(\text{Conv}_{n \times n}(\mathbf{X}))]) \quad (4)$$

In the equation, $\mathbf{Z}$ and $\mathbf{X}$ denote the output and input of the dilation context block, respectively. $\text{Conv}_{x \times x, d_y}$ represents the dilated convolution layer with the kernel size of $x \times x$ and the dilation rate of y .

III. EXPERIMENTS AND ANALYSIS

To evaluate the performance of the proposed APAFNet, the ablation studies and comparison experiments are conducted in this section.

A. Experimental Settings

1) *Dataset*: The experiments are conducted on the NUAA-SIRST dataset [17] and theIRSTD-1k dataset [10]. The NUAA-SIRST contains 427 infrared images and theIRSTD-1k contains 1000 infrared images. Both datasets are divided into training, validation, and test sets with a ratio of 50:30:20. To stack the images of various sizes into a batch, each image is resized to 512×512 and cropped into 480×480 during the training.

2) *Training Settings*: The experiment is totally completed by PyTorch on an NVIDIA GTX 3060 GPU. During the training process, we use Soft-IOU [20] as the loss function and Adagrad as the optimizer. The batch size, initial learning rate, and weight decay are set to 8, 0.05, and 1×10^{-4} , respectively. As the training goes on, the learning rate declines under a polynomial strategy, and the network is trained for 400 epochs.

3) *Evaluation Criteria*: To assess the performance of the proposed APAFNet, we use pixel-level and target-level metrics as the evaluation criteria. For pixel-level metrics, we use

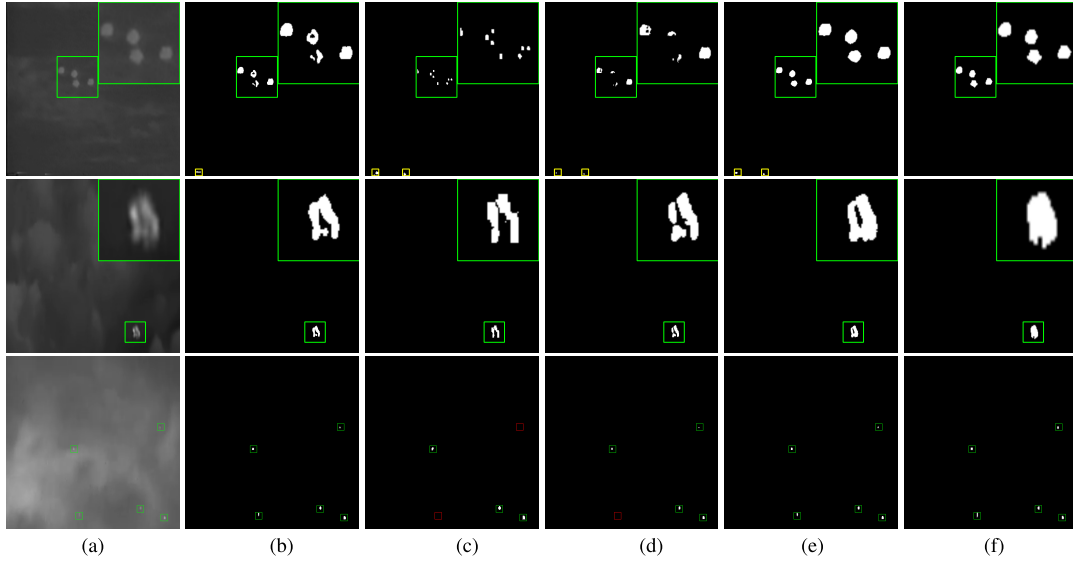


Fig. 4. Some examples of comparison between APAFNet and other methods on the NUAA-SIRST dataset. Green, yellow, and red boxes are used to denote correct predictions, false alarms, and missing detections, respectively. Especially in the first two rows, we enlarge the targets to present more details of the examples. (a) Input. (b) IPI. (c) PSTNN. (d) ACM. (e) APAFNet (ours). (f) Ground truth.

TABLE II
ABLATION STUDY ON THE IMPACT OF THE PATCH SCHEME

Patch scheme	IoU	nIoU	Pd	Fa($\times 10^{-6}$)
GAP only	0.731	0.721	0.963	9.33
2×2	0.758	0.760	0.991	18.4
3×3	0.740	0.739	0.936	10.4
4×4	0.765	0.762	0.991	8.78
5×5	0.754	0.747	0.945	21.2
6×6	0.737	0.732	0.991	14.1
10×10	0.753	0.751	0.982	27.1

Intersection over Union (IoU) and Normalized nIoU to measure the total and average overlap between the predicted results and the ground truth

$$\text{IoU} = TP / (T + P - TP) \quad (5)$$

$$\text{nIoU} = \frac{1}{N} \sum_i TP[i] / (T[i] + P[i] - TP[i]) \quad (6)$$

where N means the number of targets, and T , P , and TP denotes ground-truth pixels, predicted pixels, and true positive pixels, respectively.

We use the probability of detection (P_d) and false-alarm rate (F_a) as target-level criteria. P_d refers to the ratio of corrected predicted targets T_{correct} over all targets T_{all} , while F_a presents the ratio of false alarm pixels P_{false} over all image pixels P_{all} .

$$P_d = T_{\text{correct}} / T_{\text{all}}, F_a = P_{\text{false}} / P_{\text{all}}. \quad (7)$$

B. Ablation Study

In this section, ablation studies are conducted to verify the positive impact of each component on the whole network. Moreover, we analyze how the components benefit the network by slightly adjusting the structure inside the components. During the ablation study, ACM [17] with the same backbone network is taken as the baseline.

TABLE III
ABLATION STUDY ON THE IMPACT OF THE DILATION SCHEME

Dilation rates	IoU	nIoU	Pd	Fa($\times 10^{-6}$)
-	0.731	0.721	0.963	9.33
1	0.729	0.758	0.706	19.0
1, 3	0.720	0.723	0.963	35.4
3, 5	0.708	0.711	0.982	21.2
1, 3, 5	0.752	0.747	0.972	27.8
1, 3, 5, 7	0.734	0.748	0.963	7.72

TABLE IV
ABLATION STUDY ON THE COMPONENTS

APAF Module	DCB	IoU	nIoU	Pd	Fa($\times 10^{-6}$)
-	-	0.731	0.721	0.963	9.33
✓	-	0.765	0.762	0.991	8.78
-	✓	0.752	0.747	0.972	27.8
✓	✓	0.784	0.765	0.991	5.46

1) *Study on Patch Scheme of the APAF Module:* The output size of the adaptive pooling layer, named patch scheme for convenience, denotes how we divide the image into patches. Therefore, a dense patch scheme leads to more patches with small sizes, while a sparse one generates fewer patches with large sizes. Experiments are conducted to explore the relationship between the performance of the network and the patch scheme. The results are shown in Table II, which indicates that 4×4 achieves the best. Too dense a patch scheme shrinks the region for our patch channel attention and thus raises some pixels that are salient in a quite small region as potential targets, increasing the false alarm rates. On the contrary, too sparse a patch scheme suffers a similar problem as single utilization of global pooling, ending up with more miss detection.

2) *Study on Dilation Scheme of Dilation Context Block:* Several combinations of dilation rates are applied in the experiments and the results are shown in Table III, which implies that more local information is acquired with the increasing

TABLE V
COMPARISON WITH OTHER METHODS

Dataset	Metrics	Tophat [4]	MaxMedian [3]	IPI [7]	RIPT [8]	PSTNN [9]	MDvsFA [16]	ACM [17]	ALCNet [15]	APAFNet
NUAA-SIRST	IoU	0.0714	0.0417	0.257	0.110	0.224	0.603	0.723	0.743	0.784
	nIoU	0.0520	0.0215	0.246	0.102	0.224	0.583	0.714	0.731	0.765
	Pd	0.798	0.692	0.856	0.791	0.780	0.894	0.963	0.973	0.991
	Fa($\times 10^{-6}$)	1012	55.3	11.5	22.6	29.1	56.4	9.32	20.2	5.46
IRSTD-1k	IoU	0.101	0.0700	0.279	0.141	0.246	0.495	0.610	0.620	0.692
	nIoU	0.0744	0.0305	0.205	0.0809	0.179	0.474	0.580	0.596	0.635
	Pd	0.751	0.652	0.814	0.776	0.720	0.821	0.906	0.922	0.926
	Fa($\times 10^{-6}$)	1432	59.7	16.2	28.3	35.3	80.3	21.8	31.6	9.01

dilation rates. However, when the block takes too remote pixels into account, more interference will be brought, resulting in a misunderstanding of the context information.

3) *Ablation Study on the Components*: In order to investigate the effect of each component in the network, we collect the results of the network using the APAF module only, using the dilation context block only, and using both. Also, we compare the results with the baseline, which are presented in Table IV. As the table indicates, the APAF module and the dilation context block both bring a boost to the network and the network performs even better when adopting both.

C. Comparison With State-of-the-Art Methods

We compare the performance of the proposed network on NUAA-SIRST and ISTD-1k with other state-of-the-art methods, including traditional approaches and deep-learning-based methods. The results of the comparisons are demonstrated in Table V, which suggests that the proposed APAFNet apparently achieves better performance than other methods. Some examples of the comparisons are illustrated Fig. 4. As the figure indicates, the proposed APAFNet is able to detect quite dim and small targets without an increase in false alarms. Moreover, the proposed APAFNet has the capability of precisely segmenting the edge of the targets.

IV. CONCLUSION

In this letter, an APAFNet for SIRST detection is proposed. By using an APAF module, the network modulates low-level spatial details and high-level semantic information in both local and global scenes. Furthermore, a dilation context block is raised to obtain more local messages. Experimental results show that the APAFNet can detect dim and small targets under complex backgrounds and different interference, outperforming other existing methods.

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